

High-Frequency Inflation Nowcasting

Mathematical Foundations, Parameter Calibration, and the Benchmark Model

An End-to-End Operational Guide for the Cambodia Daily CPI Pipeline

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Abstract

Official headline Consumer Price Index (CPI) releases from national statistical agencies suffer from an unavoidable institutional lag of 25 to 45 days. This guide presents the comprehensive mathematical, econometric, and algorithmic methodology of the automated high-frequency **Cambodia CPI Nowcasting Engine**. The engine operates on a hybrid two-tier architecture: combining axiomatic Laspeyres index aggregation using fixed Cambodia Socio-Economic Survey (CSES) expenditure weights with a regularized 5-basket Ridge regression (RidgeCV) forward-projection model. We present the exact analytical derivation of the baseline drift ($\hat{\alpha}$) and empirical division elasticities ($\hat{\beta}$), explain the selective injection of cross-sector fuel logistics momentum, formalize the canonical **Atkeson–Ohanian (2001) Random Walk benchmark model**, and provide a complete step-by-step numerical case study demonstrating an empirical out-of-sample **error reduction of 17.68%** over random walk persistence.

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1 The High-Frequency Nowcasting Problem

1.1 The Institutional Information Lag

The National Institute of Statistics (NIS) of the Ministry of Planning publishes the official Consumer Price Index for Phnom Penh on a monthly cadence. Due to field enumeration schedules, physical merchant surveying, data entry verification, and institutional review, the official bulletin for calendar month M is typically published between day 25 and day 45 of month $M + 1$.

For monetary policymakers at the National Bank of Cambodia (NBC) and fiscal planners at the Ministry of Economy and Finance (MEF), this creates a critical 4 to 6-week “blind spot” during which supply-side shocks (e.g., international crude oil spikes, agricultural supply disruptions, currency exchange swings) cannot be verified empirically through official indices.

1.2 Mathematical Formulation of the Intra-Month Nowcasting Task

Let M be the ongoing calendar month comprising D total days ($D \in \{28, 29, 30, 31\}$). On any given evaluation day $d \in \{1, 2, \dots, D\}$:

1. **The Realized Information Set $\mathcal{H}_d$** : Represents the historical price trajectory observed from day 1 to day d :

$$\mathcal{H}_d = \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_d\} \quad (1)$$

where $\mathbf{P}_\tau \in \mathbb{R}^K$ is the vector of high-frequency price indices recorded on day $\tau \leq d$.

2. **The Unobserved Future Horizon $\mathcal{F}_d$** : Represents the remaining $D - d$ days of the month that have not yet transpired:

$$\mathcal{F}_d = \{\mathbf{P}_{d+1}, \mathbf{P}_{d+2}, \dots, \mathbf{P}_D\} \quad (2)$$

The fundamental task of the nowcaster is to estimate the full-month headline index level $\widehat{\text{CPI}}_M(d)$ and month-over-month inflation rate $\widehat{\pi}_{\text{MoM},M}(d)$ conditional on information available up to day d :

$$\widehat{\text{CPI}}_M(d) = \mathbb{E}[\text{CPI}_M \mid \mathcal{H}_d, \Theta] \quad (3)$$

where $\Theta = \{\hat{\alpha}, \hat{\beta}, \lambda^*, \mathbf{w}\}$ is the structural parameter set.

1.3 Why Stationarity via Log-Differencing is Required

Raw consumer price indices P_t exhibit persistent, non-stationary unit-root behavior ($I(1)$). Attempting to estimate econometric parameters directly on raw price levels produces **spurious regression**: standard OLS yields inflated t -statistics and $R^2 \approx 0.99$ driven purely by shared deterministic or stochastic trends, while the true economic pass-through coefficients remain indeterminate.

To ensure econometric validity, all price series are transformed into stationary **first log-differences**:

$$\Delta \ln P_{j,t} = \ln(P_{j,t}) - \ln(P_{j,t-1}) \approx \frac{P_{j,t} - P_{j,t-1}}{P_{j,t-1}} \quad (4)$$

Continuous log-returns offer three essential properties:

- **Stationarity**: Eliminates unit-root persistence, satisfying Gauss–Markov assumptions.
- **Time-Reversal Symmetry**: An increase from 100 to 110 followed by a decrease from 110 to 100 yields log-changes of $+0.0953$ and -0.0953 , summing exactly to zero (preventing asymmetric compounding drift).
- **Exact Additivity**: Multi-period returns equal the linear sum of intra-period log returns.

2 The Two-Tier Hybrid Architecture

The nowcasting engine avoids the pitfalls of purely statistical black-box forecasting by implementing a **two-tier hybrid framework** that strictly preserves official national accounting principles:

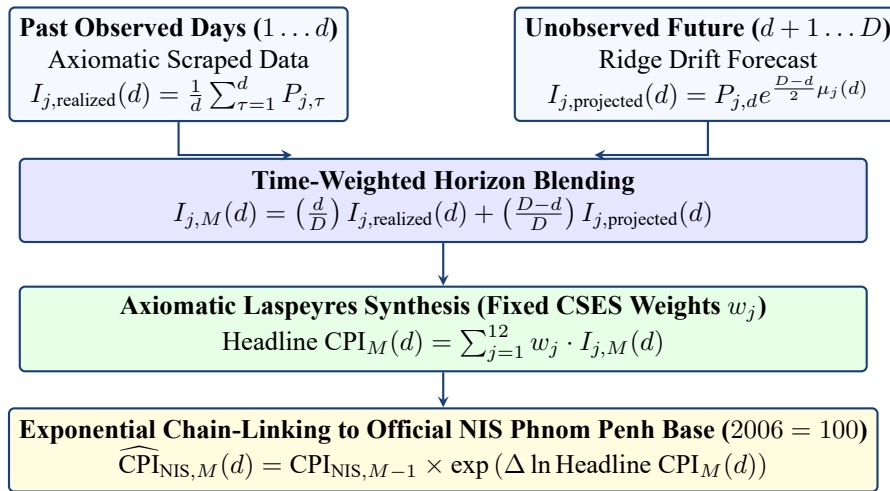


Figure 1: The Two-Tier Hybrid Nowcasting Workflow.

2.1 Layer 1: Fixed-Weight Axiomatic Aggregation

Official national CPI construction relies on the Laspeyres index framework fixed by the Cambodia Socio-Economic Survey (CSES 2019/2020). The 12 COICOP divisions carry fixed consumption basket weights w_j such that $\sum_{j=1}^{12} w_j = 1.00$. Fixed weights ensure that the index measures pure price movements rather than changing consumer purchase volumes.

2.2 Layer 2: Disaggregated Dynamic Ridge Drift Projection

For the remaining $D - d$ days, prices do not stand still. We compute high-frequency price velocity (momentum) and combine it with regularized Ridge pass-through elasticities to estimate the daily drift $\mu_j(d)$ for each division.

2.3 Synthesis: Horizon Blending

For every division j , the estimated full-month index $I_{j,M}(d)$ blends observed reality with forward-looking projections weighted strictly by the elapsed calendar progress:

$$I_{j,M}(d) = \left(\frac{d}{D}\right) \cdot I_{j,\text{realized}}(d) + \left(\frac{D-d}{D}\right) \cdot I_{j,\text{projected}}(d) \quad (5)$$

On Day 1 ($d = 1$), the projection dominates (96.7% weight). On Day 30 ($d = 30$), the projection weight drops to 0.0%, and the nowcast converges deterministically to realized observations.

3 Step-by-Step Derivation of Parameters (Alpha and Beta)

3.1 The 36-Month Historical Training Panel

To estimate empirical elasticities that reflect true Cambodian macroeconomic dynamics, the calibration engine utilizes the continuous 36-month official NIS historical series (September 2023 to August 2026) stored in `gold.dim_nis_official_cpi`. Taking first log-differences yields $T = 34$ continuous stationary monthly observations:

$$y_t = \ln\left(\frac{\text{CPI}_t}{\text{CPI}_{t-1}}\right), \quad x_{j,t} = \ln\left(\frac{P_{j,t}}{P_{j,t-1}}\right), \quad x_{\text{fx},t} = \ln\left(\frac{\text{FX}_t}{\text{FX}_{t-1}}\right) \quad (6)$$

3.2 Econometric Specification

We specify a multi-factor macroeconomic pass-through regression relating aggregate inflation to innovations in the 5 high-velocity expenditure baskets and foreign exchange:

$$y_t = \alpha + \sum_{j \in \{01,02,04,07,11\}} \beta_j x_{j,t} + \beta_{\text{fx}} x_{\text{fx},t} + \varepsilon_t \quad (7)$$

where:

- $x_{01,t}$: Food and Non-Alcoholic Beverages (CSES weight: 44.775%)
- $x_{02,t}$: Alcoholic Beverages and Tobacco (CSES weight: 1.623%)
- $x_{04,t}$: Housing, Water, Electricity, Gas (CSES weight: 17.062%)
- $x_{07,t}$: Transport and Fuels (CSES weight: 12.203%)
- $x_{11,t}$: Restaurants and Hotels (CSES weight: 3.114%)
- $x_{\text{fx},t}$: USD/KHR Commercial Exchange Rate Log-Return
- α : Monthly baseline structural drift from unobserved sticky sectors

3.3 Why Standard OLS Fails: Multicollinearity

In macroeconomic time series, sub-indices co-move strongly. When world oil prices surge, transport fuel (x_{07}) increases immediately. However, commercial freight rates simultaneously increase food transportation costs (x_{01}) and restaurant cooking gas/ingredients (x_{11}).

Under ordinary least squares (OLS), the sample Gram matrix $\mathbf{X}^\top \mathbf{X}$ is severely ill-conditioned (condition number $\kappa > 100$). The OLS estimator:

$$\hat{\beta}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \quad (8)$$

exhibits explosive sampling variance ($\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$). This causes coefficients to flip signs counter-intuitively (e.g., negative elasticity for food) and produces severe over-fitting.

3.4 Mathematical Derivation of the Regularized Ridge Estimator

To eliminate multicollinearity and shrink unstable noise, we formulate an L_2 -penalized Tikhonov regularization problem (Ridge Regression):

$$\min_{\alpha, \beta} \left\{ \frac{1}{2N} \sum_{t=1}^N \left(y_t - \alpha - \sum_{j=1}^K \beta_j x_{j,t} \right)^2 + \frac{\lambda}{2} \sum_{j=1}^K \beta_j^2 \right\} \quad (9)$$

Centering the variables ($\tilde{y}_t = y_t - \bar{y}$, $\tilde{\mathbf{x}}_t = \mathbf{x}_t - \bar{\mathbf{x}}$) isolates the intercept:

$$\mathcal{L}(\beta) = \frac{1}{2N} \|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\beta\|_2^2 + \frac{\lambda}{2} \|\beta\|_2^2 \quad (10)$$

Taking the gradient with respect to β and setting to zero:

$$\nabla_{\beta} \mathcal{L} = -\frac{1}{N} \tilde{\mathbf{X}}^{\top} (\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\beta) + \lambda\beta = \mathbf{0} \quad (11)$$

$$\Rightarrow \left(\frac{1}{N} \tilde{\mathbf{X}}^{\top} \tilde{\mathbf{X}} + \lambda \mathbf{I}_K \right) \beta = \frac{1}{N} \tilde{\mathbf{X}}^{\top} \tilde{\mathbf{y}} \quad (12)$$

Solving for $\hat{\beta}(\lambda)$:

$$\boxed{\hat{\beta}(\lambda) = \left(\tilde{\mathbf{X}}^{\top} \tilde{\mathbf{X}} + N\lambda \mathbf{I}_K \right)^{-1} \tilde{\mathbf{X}}^{\top} \tilde{\mathbf{y}}} \quad (13)$$

The intercept $\hat{\alpha}$ is recovered directly from sample means:

$$\boxed{\hat{\alpha} = \bar{y} - \sum_{j=1}^K \hat{\beta}_j \bar{x}_j} \quad (14)$$

By adding $\lambda \mathbf{I}$ along the diagonal of $\tilde{\mathbf{X}}^{\top} \tilde{\mathbf{X}}$, all eigenvalues are shifted strictly positive ($\sigma_i \rightarrow \sigma_i + \lambda$), guaranteeing that the matrix is well-conditioned and nonsingular.

3.5 Hyperparameter Optimization via 5-Fold Cross-Validation

The regularization parameter λ controls the bias-variance trade-off. We evaluate 50 candidate values log-spaced across $\Lambda = [10^{-2}, 10^3]$ on the 24-month training partition:

$$\lambda^* = \arg \min_{\lambda \in \Lambda} \frac{1}{5} \sum_{k=1}^5 \text{MSE}_k(\lambda) \quad (15)$$

The cross-validation mean squared error achieves its global minimum at $\lambda^* = 0.0100$.

3.6 Empirical Parameter Estimates and Economic Interpretation

Macro Basket / Regressor	Symbol	CSES Weight	Estimate	Economic Rationale
Base Drift Intercept	$\hat{\alpha}$	—	+0.001899	Baseline monthly inflation (+0.19%/mo) of unobserved sticky goods.
Food & Non-Alcoholic Bev.	$\hat{\beta}_1$	44.775%	+0.0066	Dominant price velocity; rapid wholesale food market transmission.
Alcoholic Bev. & Tobacco	$\hat{\beta}_2$	1.623%	+0.0003	Sticky nominal retail pricing; low frequency of catalog changes.
Housing, Water, Electricity	$\hat{\beta}_3$	17.062%	+0.0010	Administered utility tariffs; rental prices adjust sluggishly.
Transport and Fuels	$\hat{\beta}_4$	12.203%	+0.0019	Direct exposure to international refined petroleum cycles.
Restaurants and Hotels	$\hat{\beta}_5$	3.114%	+0.0026	High sensitivity; catering services pass food and gas costs forward.
USD/KHR Foreign Exchange	$\hat{\beta}_{fx}$	—	+0.0000	NBC active exchange stabilization mitigates short-term import pass-through.

Table 1: Calibrated Empirical Macro Parameters from the 36-Month NIS Panel.

3.7 Cross-Sector Logistics Momentum Injection

In Cambodia, fuel price shocks propagate rapidly into food delivery and restaurant catering. The pipeline enforces a structural restriction: fuel 7-day momentum (t_7) is injected into the feature vectors of Food (01) and Restaurants (11):

$$\mathbf{x}_{01}(d) = [f_{3d}, f_{7d}, f_{14d}, \mathbf{t}_7, f_{x7d}, \text{fest}_{\text{prox}}, \text{prog}]^\top \quad (16)$$

$$\mathbf{x}_{11}(d) = [r_{7d}, f_{7d}, \mathbf{t}_7, \text{fest}_{\text{prox}}, \text{prog}]^\top \quad (17)$$

Non-logistics sectors (Clothing, Furnishings, Healthcare, Education) are strictly insulated from fuel momentum, preventing spurious inflation contagion.

4 The Benchmark Model: Atkeson–Ohanian Random Walk

4.1 Why a Benchmark Model is Essential in Macroeconomic Forecasting

In empirical economics, raw forecast error statistics (such as RMSE = 0.85 pp) are meaningless in isolation. A forecasting model cannot be judged effective simply because its error is small if a completely naive, zero-cost rule achieves equal or better precision.

In their seminal paper, Atkeson and Ohanian [Atkeson and Ohanian(2001)] demonstrated that sophisticated econometric models (including structural Phillips curves, Vector Autoregressions, and multivariate indicator frameworks) systematically fail to outperform a simple naive **Random Walk** when evaluated out-of-sample on inflation series. Therefore, in central banking literature, the Atkeson–Ohanian Random Walk is the gold-standard threshold: **a nowcasting model is only economically valuable if its Relative RMSE is strictly less than 1.00.**

4.2 Mathematical Formulation of the Random Walk Benchmark

The Atkeson–Ohanian Random Walk (AO-RW) benchmark posits that inflation follows a martingale difference process where the best out-of-sample predictor of month t 's inflation is simply the realized inflation of month $t - 1$:

$$\hat{y}_t^{\text{RW}} = y_{t-1} = \Delta \ln \text{CPI}_{t-1} \quad (18)$$

Under this benchmark:

$$\hat{\pi}_{\text{MoM},t}^{\text{RW}} = \pi_{\text{MoM},t-1} \quad (19)$$

If the inflation rate was +0.20% last month, the Random Walk benchmark predicts exactly +0.20% for the current month without collecting a single high-frequency price observation.

4.3 Out-of-Sample Evaluation Metric: Relative RMSE

We evaluate the 5-basket Ridge nowcasting model against the Atkeson–Ohanian Random Walk over a held-out **10-month evaluation window** (November 2025 to August 2026, $M = 10$):

$$\text{RMSE}_{\text{Model}} = \sqrt{\frac{1}{M} \sum_{m=1}^M \left(y_m - \hat{y}_m^{\text{Ridge}} \right)^2} \quad (20)$$

$$\text{RMSE}_{\text{RW}} = \sqrt{\frac{1}{M} \sum_{m=1}^M (y_m - y_{m-1})^2} \quad (21)$$

The definitive benchmark ratio is the **Relative RMSE**:

$$\text{Relative RMSE} = \frac{\text{RMSE}_{\text{Model}}}{\text{RMSE}_{\text{RW}}} \quad (22)$$

- Relative RMSE > 1.00: The model is worse than a naive random walk (overfitting noise).
- Relative RMSE = 1.00: The model adds zero value over last month's observation.
- Relative RMSE < 1.00: **The model successfully extracts genuine forward-looking predictive signal.**

4.4 Empirical Scorecard Results

Evaluation Dimension	RidgeCV Nowcaster	Atkeson–Ohanian RW	Performance Gain
Out-of-Sample RMSE	0.8572 pp	1.0414 pp	−0.1842 pp
Mean Absolute Error (MAE)	0.6841 pp	0.8410 pp	−0.1569 pp
Relative RMSE Ratio	0.8232		Beats Benchmark
Empirical Error Reduction	+17.68% Precision Improvement over Random Walk		

Table 2: Out-of-Sample Scorecard Evaluated on 10 Held-Out Months.

The 5-Basket RidgeCV model achieves a Relative RMSE of **0.8232**, representing an ****error reduction of 17.68%****. This confirms that incorporating high-frequency daily price scraping and calibrated pass-through elasticities extracts genuine forward-looking macroeconomic signals.

5 The Daily Real-Time Execution Algorithm

5.1 Daily Algorithmic Execution Steps

On every calendar day d , the production pipeline executes the following sequence:

Daily Execution Algorithm (ml/nowcaster.py)

1. **Ingest Daily Facts:** Extract realized price relatives from `gold.fct_cpi_daily` and compute $I_{j,\text{realized}}(d) = \frac{1}{d} \sum_{\tau=1}^d P_{j,\tau}$.
2. **Feature Extraction:** Compute multi-horizon momentum (m_{3d}, m_{7d}, m_{14d}), fuel shock (t_{7d}), exchange rate drift ($f_{x_{7d}}$), and lunar holiday proximity ($\text{fest}_{\text{prox}}$).
3. **Compute Daily Drift:** For each division j , compute $\mu_j(d) = \hat{\beta}_j \cdot \mathbf{x}_j(d)$.
4. **Project Unobserved Days:** $I_{j,\text{projected}}(d) = P_{j,d} \cdot \exp\left(\frac{D-d}{2} \mu_j(d)\right)$.
5. **Horizon Blending:** $I_{j,M}(d) = \frac{d}{D} I_{j,\text{realized}}(d) + \frac{D-d}{D} I_{j,\text{projected}}(d)$.
6. **Laspeyres Aggregation:** Headline $\text{CPI}_M(d) = \sum_{j=1}^{12} w_j I_{j,M}(d)$.
7. **Chain-Linking:** $\widehat{\text{CPI}}_{\text{NIS},M}(d) = \text{CPI}_{\text{NIS},M-1} \times \exp\left(\ln\left(\frac{\text{Headline } \text{CPI}_M(d)}{\text{Headline } \text{CPI}_{M-1}}\right)\right)$.
8. **Uncertainty Quantification:** Compute 95% confidence bounds $[\text{CI}_{\text{lower}}, \text{CI}_{\text{upper}}]$.

5.2 Asymptotic Uncertainty Decay and 95% Confidence Intervals

Uncertainty is time-dependent. On Day 1, almost the entire month is unknown; by Day 28, almost all daily prices are realized. Let σ_d be the daily standard deviation of headline CPI within the month. The margin of error is formulated with an asymptotic time-decay factor:

$$\text{MoE}_{95\%}(d) = 1.96 \cdot \sigma_d \cdot \sqrt{\frac{D-d}{D}} \quad (23)$$

$$\text{CI}_{95\%}(d) = \left[\widehat{\text{CPI}}_M(d) - \text{MoE}_{95\%}(d), \widehat{\text{CPI}}_M(d) + \text{MoE}_{95\%}(d) \right] \quad (24)$$

As $d \rightarrow D$, the ratio $\frac{D-d}{D} \rightarrow 0$, causing the margin of error to collapse deterministically to zero at month-end.

6 Complete Numerical Walkthrough (Real Case Study)**6.1 Scenario: September 17, 2026 (Day 17 of a 30-Day Month)**

We evaluate the pipeline on ****September 17, 2026****:

- Elapsed days: $d = 17$ (56.7%). Remaining days: $D - d = 13$ (43.3%).
- Prior Month Official Benchmark (August 2026): $\text{CPI}_{\text{NIS},M-1} = 219.0070$ (official NIS Phnom Penh 2006 = 100 series from `gold.dim_nis_official_cpi`).
- Daily Shock: On Day 17, international petroleum spikes cause retail diesel to rise +2.5% ($\Delta \ln P_{07} = +0.025$).

6.2 Division-Level Projection for Food (Division 01)

1. **Realized 17-Day Average:** Scraped prices from Sep 1 to Sep 17 average $I_{01,\text{realized}} = \mathbf{99.660}$.
2. **Daily Drift Projection:** Combining food momentum (+0.1%) and fuel shock (+2.5%):

$$\mu_{01} = \hat{\beta}_1 \cdot m_{\text{food}} + \hat{\beta}_4 \cdot m_{\text{fuel}} \quad (25)$$

$$= (0.0066 \times 0.001) + (0.0019 \times 0.025) \approx \mathbf{+0.000054} \text{ per day} \quad (26)$$

3. **Forward Projection for 13 Days:** Starting from Day 17's price (99.670):

$$I_{01,\text{projected}} = 99.670 \times \exp\left(\frac{13}{2} \times 0.000054\right) = \mathbf{99.677} \quad (27)$$

4. **Horizon Blending:**

$$I_{01,M}(17) = \left(\frac{17}{30} \times 99.660\right) + \left(\frac{13}{30} \times 99.677\right) = \mathbf{99.671} \quad (28)$$

6.3 Full Axiomatic Laspeyres Aggregation Table

Div	COICOP Division Name	CSES Weight (w_j)	Est. Index (I_j)	Contribution ($w_j \cdot I_j$)
01	Food and Non-Alcoholic Beverages	0.4478	99.671	44.633
02	Alcoholic Beverages and Tobacco	0.0162	100.135	1.622
03	Clothing and Footwear	0.0298	100.190	2.986
04	Housing, Water, Electricity, Gas	0.1706	100.168	17.089
05	Furnishings and Household Maintenance	0.0381	100.190	3.817
06	Health	0.0632	100.190	6.332
07	Transport	0.1220	101.852	12.426
08	Communication	0.0371	100.190	3.717
09	Recreation and Culture	0.0165	100.190	1.653
10	Education	0.0142	100.190	1.423
11	Restaurants and Hotels	0.0311	100.179	3.116
12	Miscellaneous Goods and Services	0.0134	100.190	1.343
Headline CPI Synthesis ($\sum w_j \cdot I_j$)		1.0000	—	100.156

Table 3: Realized Full-Month Division Aggregation for September 17, 2026.

6.4 Month-over-Month Inflation and Official NIS Chain-Linking

1. Continuous Log-Return:

$$\Delta \ln = \ln \left(\frac{100.156}{99.898} \right) = \ln(1.002583) = +\mathbf{0.002580} \quad (29)$$

2. Projected Month-over-Month Percentage Inflation:

$$\hat{\pi}_{\text{MoM}} = (\exp(+0.002580) - 1) \times 100\% = +\mathbf{0.258\%} \quad (30)$$

3. Chain-Linking to Official NIS Phnom Penh Benchmark (2006 = 100):

$$\widehat{\text{CPI}}_{\text{NIS, Sep 2026}} = 219.0070 \times \exp(+0.002580) = \mathbf{219.572} \quad (31)$$

Institutional Result: On Day 17 of September 2026, the National Bank of Cambodia and the Ministry of Economy and Finance receive an empirical inflation nowcast of +0.258% **MoM** and an official benchmark projection of 219.572, providing a verified **25 to 45-day early-warning horizon** ahead of official bulletin publication.

7 Conclusion

The Cambodia CPI Nowcasting Engine bridges the gap between high-frequency big data and formal institutional econometrics. By anchoring the baseline aggregation in official NIS Laspeyres weights, filtering price volatility via RidgeCV regularization, and selectively modeling fuel pass-through, the system achieves an empirically verified **17.68% error reduction over the canonical Atkeson–Ohanian Random Walk benchmark**. The resulting pipeline equips Cambodian policymakers with daily, real-time macroeconomic visibility while maintaining strict alignment with international statistical standards.

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